

Lightweight Residual Networks for Small-Scale Image Classification

Abstract

We propose a lightweight residual network for small-scale image classification. Our method achieves 94.3% accuracy on CIFAR-10 v2.1 with only 2.7M parameters. We describe the training setup in detail to support reproducibility.

1. Introduction

Deep residual networks have achieved remarkable success in image classification. However, reproducing published results remains a challenge due to incomplete training details.

2. Related Work

ResNet and its variants form the backbone of modern image classification systems. Wide ResNets and DenseNets further improve accuracy at the cost of additional parameters.

3. Method

Our network follows the ResNet-18 architecture with width multiplier 0.5. We use input size 32x32 and dropout 0.1 in the classifier head.

4. Training Setup

We train all models on CIFAR-10 v2.1 using the Adam optimizer. The learning rate is set to $1e-4$ with weight decay $5e-4$. We use a batch size of 32 and train for 100 epochs. The random seed is fixed to 42 for all experiments. We apply a cosine annealing learning rate scheduler.

5. Experiments

Our method achieves 94.3% top-1 accuracy on the CIFAR-10 test set. The final test loss is 0.21. All results are averaged over 3 runs with the fixed seed.